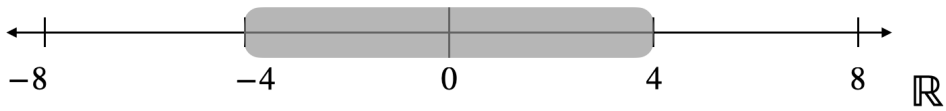
A Note on the Solutions: This solution sheet is intended for you verify correctness of your work during the tutorial. The solutions here do not necessarily meet the standards for exposition expected on assignment submissions.

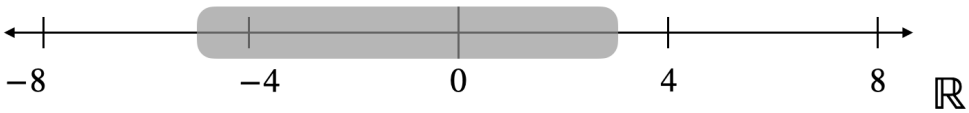
1. **The Geometry of Inequalities**

   For each inequality, find all values of x that make the inequality true. Draw a picture of your solution on the real line.

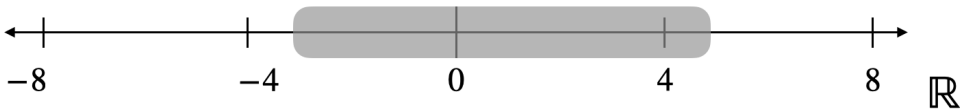
   (a) $|x| < 4$.



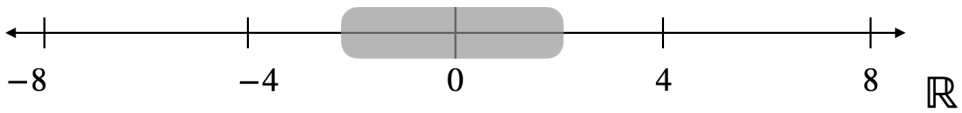
   (b) $|x + 1| < 4$



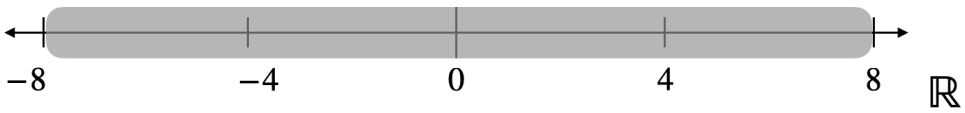
   (c) $|x - 1| < 4$



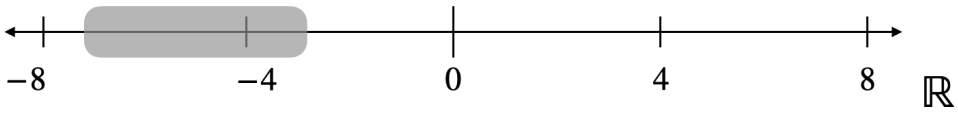
   (d) $|2x| < 4$



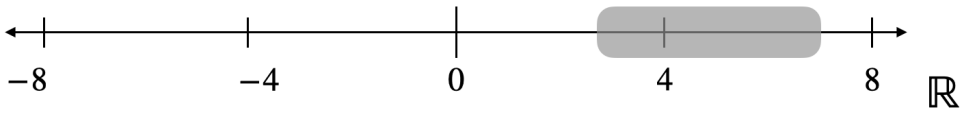
   (e) $|x/2| < 4$



   (f) $|2x + 10| < 4$



   (g) $| - 2x + 10| < 4$



2. **Transformations.** At some point in the past you have learned about *transformations* of polynomial functions. For example, given a polynomial $f(x)$ we can draw conclusions about the *shape* of $f(ax+b)+c$ using words like *stretch* or *translate*. When you compare your drawings of the solutions in the previous question, perhaps we can see the same sort of behaviour.

   Let $a, b, c \in \mathbb{R}$, with $a, c > 0$

   (a) What *transformation* turns the set of solutions for $|x| < c$ into the set of solutions for $|x+b| < c$?

   Translation

   (b) What *transformation* turns the set of solutions for $|x| < c$ into the set of solutions for $|ax| < c$?

   Stretch/compress

   (c) What *transformation* turns the set of solutions for $|ax+b| < c$ into the set of solutions $|-ax+b| < c$?

   Reflection

   (d) What *transformation* turns the set of solutions for $|ax+b| < c$ into the set of solutions for $|ax+b| \geq c$. Can you express this *transformation* using the language of set theory?

   Inversion

   This can be viewed as set complement. If A is the set of solutions for $|ax+b| < c$, then $\mathbb{R} \setminus A$ is the set of solutions for $|ax+b| \geq c$.

3. **The Triangle Inequality**

   Give an example of real numbers x, y such that the triangle inequality holds as (i) an equality; (ii) a *strict* inequality.

   Solution: For $x = 1, y = 2$ we have:

$$|x+y| = |3| = 3 = 1+2 = |1|+|2| = |x|+|y|$$

   For $x = 1$ and $y = -1$ we have:

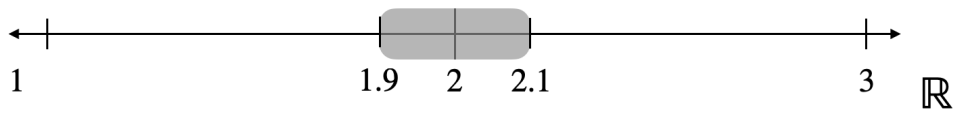
$$|x+y| = |0| = 0 < 1+1 = |1|+|-1| = |x|+|y|$$

4. **Neighbourhoods**

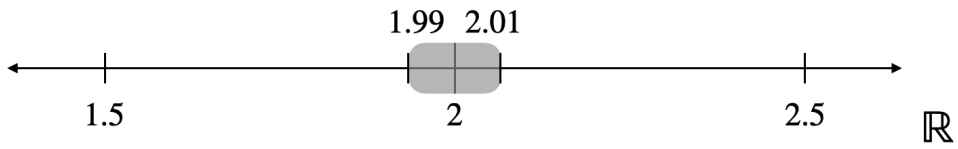
   Consider the following definition:

   Definition. *Let $r, \ell, \epsilon \in \mathbb{R}$ with $\epsilon > 0$. We say <u>ℓ is in the ϵ-neighbourhood of r</u> when $|\ell - r| < \epsilon$*

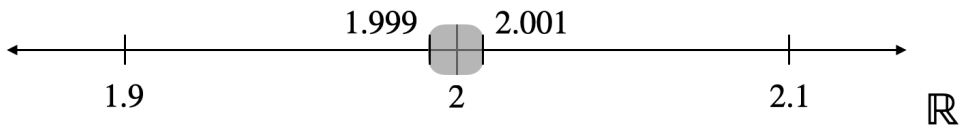
   (a) For which values of ℓ is ℓ in 0.1-neighbourhood of 2? Draw a picture of the ϵ-neighbourhood of r on the real line to help your understanding.



   (b) For which values of ℓ is ℓ in 0.01-neighbourhood of 2? Draw a picture of the ϵ-neighbourhood of r on the real line to help your understanding.



   (c) For which values of ℓ is ℓ in 0.001-neighbourhood of 2? Draw a picture of the ϵ-neighbourhood of r on the real line to help your understanding.



(d) Let $\epsilon > 0$. Is it true that the ϵ-neighbourhood of 2 contains a real number other than 2? Justify your response by using the Archimedian Principle v3.

$\ell \in \mathbb{R}$ is in the ϵ-neighbourhood of 2 when $|\ell - 2| < \epsilon$.

By the Archiemedian Principle v3, for any $x, y \in \mathbb{R}^+$ there exists $n \in \mathbb{N}$ so that $y < nx$. Consider $x = \epsilon$ and $y = 1$. There exists $n \in \mathbb{N}$ so that $1 < n\epsilon$. In other words, $1/n < \epsilon$.

Let $\ell = 2 + 1/n$. By construction we have

$$|\ell - 2| = |2 + 1/n - 2| = 1/n < \epsilon$$

Therefore $2 + 1/n$ is in the ϵ-neighbourhood of 2.

5. **Bounding Functions**

To <u>bound</u> a function f on an interval $[a, b]$ means to find a real number K such that $|f(x)| \leq K$ for all $x \in [a, b]$.

(a) Bound the expression $|x - 3|$ on $[-2, 4]$.

Solution: As $-2 \leq x \leq 4$, we have

$$-5 \leq x - 3 \leq 1 \quad \Rightarrow \quad |x - 3| \leq |-5| = 5.$$

(b) Bound the expression

$$\frac{1}{|x - 3|}$$

for $x \in (-2, 0)$. Can it be bounded on $[-2, 4]$?

Solution: For $-2 < x < 0$,

$$-5 < x - 3 < -3 \quad \Rightarrow \quad -\frac{1}{3} < \frac{1}{x - 3} < -\frac{1}{5} \quad \Rightarrow \quad \frac{1}{|x - 3|} < \left|-\frac{1}{3}\right| = \frac{1}{3}.$$

Since $|x - 3|$ blows up near $x = 3$ and $3 \in [-2, 4]$, the expression is *not* bounded.

(c) Bound the function $f : (-2, 7) \to \mathbb{R}$ given by

$$f(x) = 2x^2 - 6x + 7.$$

Hint: Any upper bound on $|f|$ will do! You can use the triangle inequality to get a bound here.

Solution: We look for $K \geq |f(x)|$ on $-2 < x < 7$. By the triangle inequality (applied twice),

$$\left|2x^2 - 6x + 7\right| \leq |2x^2| + |6x| + |7| = 2|x|^2 + 6|x| + 7 \leq 2 \times 7^2 + 6 \times 7 + 7 = 147.$$